COMPUTER SCIENCE & ENTREPRENEURSHIP 9

UNIT 1: INTRODUCTION TO COMPUTATIONAL SYSTEMS

System Theory & Basic Concepts (Components, Environment & Communication)

Question 1: What is a system?

A system is an organized collection of interconnected and interacting components that work together to perform a specific task or achieve a common objective.

Question 2: What is an Information System?

An information system is a combination of technology, data, and people that collects, processes, stores, and shares information to enable effective decision-making.

Question 3: What are the basic components of a computer system?

A computer system consists of input devices, output devices, a central processing unit (CPU), and storage devices working together.

Question 4: Explain the basic concepts of a system.

Any system is described through its objectives, components, boundaries, and interactions with its environment, which show its regular and coordinated flow.

Question 5: What is the System Purpose?

Every system has a defined goal that it wants to achieve. All operations of the system are performed to fulfill this objective.

Question 6: State the purpose and importance of a system.

The system's purpose clarifies what it is built for. It is important because it determines the system's performance and success.

Question 7: What is the primary purpose of a computer system?

The primary purpose of a computer system is to take raw data via input, process it, and provide useful information to users.

Question 8: What are system components?

System components are its basic building blocks. Each component has a specific role contributing to the overall system functionality.

Question 9: What is the role of components in a system?

The smooth integration and proper functioning of components ensure that the system performs its specific tasks, helping identify issues and improve performance.

Question 10: What is the environment?

The environment of a system includes everything outside the system that interacts with it or directly affects its performance.

Question 11: How does the environment affect a system?

The environment provides input to and receives output from the system. It influences performance, and systems adapt themselves to environmental changes.

Question 12: What is communication?

Communication is the interaction between system components. It is the fundamental basis of a system's operation, keeping various parts linked.

Question 13: Give an example of communication in computers.

Data exchange between the CPU and RAM or hard disk, and communication with other computers via a network, are prime examples of computer communication.

Question 14: Give an example of communication in biological systems.

In the human body, the nervous system sends signals to the brain and provides commands for organs to move or function.

Question 15: Define software.

Software is a collection of programs and instructions that tells a computer what to do and how to do it. Without software, a computer is just a lifeless machine.

Question 16: What was the first computer virus and when was it created?

The first computer virus was created in 1971 named 'Creeper', a self-replicating program across networks with the message: 'I'm the creper, catch me if you can!'

System Software & Application Software

Question 17: Define system software.

System software manages computer hardware resources and provides a platform to run application software, acting as a bridge between hardware and the user.

Question 18: Give examples of operating systems.

Well-known examples include Microsoft Windows, macOS, Linux, and Android for mobile devices.

Question 19: Give examples of device drivers.

Printer drivers, graphics card drivers, and sound card drivers enable hardware to communicate with the operating system.

Question 20: Give examples of utility programs.

Antivirus software, disk cleanup tools, file managers, and backup software are key examples of utility programs.

Question 21: Define application software.

Application software is designed to help users perform specific tasks such as creating documents, surfing the internet, or playing games.

Question 22: Give examples of various application software.

Microsoft Word, Google Chrome, VLC Media Player, Photoshop, and various educational and business software.

Question 23: What is the difference between system software and application software?

System software runs the computer and controls hardware, whereas application software fulfills specific and personal tasks for the user.

Question 24: What is the difference in installation between them?

System software is usually pre-installed or installed with the operating system, while application software is installed by the user as needed.

Question 25: Why is it important to keep system software updated?

Updating system software is vital for improving security, gaining new hardware support, and eliminating system bugs.

Von Neumann Architecture & CPU

Question 26: What is Von Neumann Architecture?

Von Neumann Architecture is a computer model describing a system where computer hardware has four basic components: memory, CPU, input, and output.

Question 27: Why was it named Von Neumann and when was it built?

It is called the Von Neumann model in honor of mathematician John von Neumann, who was involved in its formulation and design during the 1940s.

Question 28: What are the basic components of Von Neumann Architecture?

Its basic components are (1) Central Processing Unit including ALU and CU, (2) Memory, (3) Input devices, and (4) Output devices.

Question 29: What is the role of memory?

Memory is the part of the computer that stores runtime data and necessary instructions to provide high-speed access to the CPU.

Question 30: What does the Central Processing Unit do?

The CPU is called the brain of the computer. It processes all instructions, performs calculations and logic decisions, and controls all hardware activities.

Question 31: What is the function of the Arithmetic and Logic Unit (ALU)?

The ALU performs all mathematical (addition, subtraction, multiplication, division) and logical (comparison and decision-making) operations.

Question 32: What is the function of the Control Unit (CU)?

The CU controls the flow of data and instructions between all parts of the computer and ensures instruction execution at all times accurately.

Input, Output & Buses

Question 33: What are Input Devices?

They enable users to input data and instructions into the computer system. Examples: keyboard, mouse, microphone. Data transfers from these devices to the CPU.

Question 34: What is the function of Output Devices?

They display results of tasks given by the user. After processing, the CPU transfers results to the monitor for visual display.

Question 35: What is the role of buses in a computer system?

Buses are communication mechanisms enabling data movement among internal components: (1) Data bus transfers data, (2) Address bus holds memory address info, (3) Control bus transmits control signals.

Stages of Von Neumann Architecture

Question 36: What is the working mechanism in Von Neumann Architecture?

Von Neumann Architecture has four operating stages: Fetching, Decoding, Execution, and Storage.

Question 37: What is Fetching?

In this stage, the CPU fetches the next instruction from memory for processing and execution.

Question 38: Which hardware components are used during fetching?

Memory, CPU, Program Counter (PC), and Instruction Register (IR) are involved.

Question 39: What is the function of the Program Counter (PC)?

The Program Counter (PC) stores the memory address of the next upcoming instruction when it is fetched and placed in the Instruction Register (IR).

Question 40: What is Decoding and what are its related components?

During decoding, the Control Unit decodes the operation code of the instruction to determine what action to perform. Its related component is the Control Unit (CU).

Question 41: What does the Control Unit do during decoding?

The Control Unit (CU) understands the instruction and notifies relevant components regarding the action and expected result.

Question 42: What is Execution?

The CPU executes the instruction. If related to arithmetic, the ALU handles it; if related to data transfer, the Control Unit handles it.

Question 43: What are the related components during execution?

The related components are the ALU and CU.

Question 44: What is Storage?

The result of a calculation is either stored in designated memory or sent to an output device. Related components are memory and output devices.

Question 45: Explain the storage process.

In this stage, results are stored in memory for later use or displayed on an output device.

Number System Types and Conversions

Question 46: Why are number systems necessary in computing?

Number systems provide the basis for data representation, storage, and processing, allowing representation of various information types.

Question 47: What role do various number systems play in computers?

They help handle and process data inside computers in binary, decimal, octal, and hexadecimal formats.

Question 48: What is the Decimal Number System?

The decimal number system is base-10, comprising digits 0 through 9. We frequently use this system in everyday life.

Question 49: What is the Binary Number System?

In binary, place values lie right to left, starting from powers of 2. Each position represents a power of 2.

Question 50: Why do computers work in binary system?

Two states (on and off) are most easily implemented across electronic systems, which is why computers operate on binary.

Question 51: What is the algorithm for decimal to binary conversion?

Divide the number by 2 and note the remainder. Repeat division until the quotient becomes zero. Read remainders bottom-up.

Question 52: What is the Octal Number System?

The octal system is base-8, meaning digits used range from 0 to 7.

Question 53: What is the method to convert octal to binary?

Each octal digit represents three binary digits because octal base is 8 and binary base is 2.

Question 54: Explain the algorithm to convert decimal to octal.

Divide the decimal number by 8, note the remainder, and repeat until the quotient is zero.

Question 55: What is the Hexadecimal Number System?

Hexadecimal number system is base-16, using digits 0 through 9 and letters A through F.

Question 56: How is hexadecimal converted to binary?

Each hexadecimal digit equals four binary bits, helping convert it to binary easily.

Question 57: What is the method to convert decimal to hexadecimal?

Divide the number by 16, convert remainder to hex representation, and repeat until quotient is zero.

Question 58: Which system was used in early computing systems?

Early computing systems used octal or hexadecimal systems to read binary code easily.

Question 59: What is the rule for converting binary to octal?

Divide binary numbers into groups of three from right to left and find the octal digit for each group.

Question 60: What if binary number bits do not form groups of three?

Zeros are added to the left side of the binary number as needed to complete groups of three.

Question 61: What are Whole Numbers?

Whole numbers are a collection of non-negative integers including zero and all positive numbers.

Question 62: What is the maximum value in a byte of whole numbers?

An 8-bit byte can hold a maximum unsigned value of 255 (when all bits are 1).

Question 63: What is the benefit of increasing the number of bytes?

Increasing bytes (e.g., 2 or 4 bytes) increases the capacity to store larger amounts of data and larger numbers.

Question 64: Define an Integer.

Integers include both positive and negative numbers. In computer programming, they are called signed integers.

Signed Integers, 2's Complement & Binary Arithmetic

Question 65: How are positive and negative identified in Signed Integers?

In signed integers, the Most Significant Bit (MSB) on the left is used as a sign bit. If MSB is 0, value is positive; if 1, value is negative.

Question 66: What is the max and min value in a signed integer?

One bit is reserved for sign, so in an 8-bit signed integer, maximum value is 127 and minimum is -128.

Question 67: What is 1's Complement?

1's complement represents negative numbers by inverting all bits (turning 0s to 1s and 1s to 0s).

Question 68: How are negative quantities stored in computers?

Negative quantities are typically stored in computers using 2's Complement form.

Question 69: What are the steps to find 2's Complement?

Steps: (1) Invert all bits (0 to 1, 1 to 0), (2) Add 1 to the resulting 1's complement.

Question 70: What is the formula and value for min negative value in 8-bit integer?

Minimum value formula is -2^(n-1). Thus, in an 8-bit integer, minimum value is -128.

Question 71: What are Binary Arithmetic operations?

Binary arithmetic operations include addition, subtraction, multiplication, and division using only two digits (0 and 1).

Question 72: Write basic rules of binary addition.

Binary addition rules: 0+0=0, 0+1=1, 1+0=1, 1+1=10 (write 0, carry 1).

Question 73: How is Binary Subtraction performed?

Binary subtraction is done by taking 2's complement of the subtrahend and adding it to the minuend.

Question 74: How is 2's complement calculated?

All bits are inverted (1's complement), then 1 is added to the resulting value.

Question 75: What are the basic steps of binary multiplication?

Binary multiplication follows standard multiplication rules. Partial products based on multiplier bits are summed together.

Question 76: How does the CPU process binary numbers?

The CPU adds, subtracts, and processes millions of binary numbers every second to execute complex instructions and programs.

Question 77: What are the four steps of Binary Division?

Division, multiplication, subtraction, and bringing down the next bit are repeated until all bits are resolved.

Question 78: What is Text Encoding?

Techniques used to represent numeric values of characters and symbols of various languages so computers can understand them.

Question 79: What is ASCII and how many characters does it represent?

ASCII stands for American Standard Code for Information Interchange. It is a 7-bit code representing 128 different characters from 0 to 127.

Question 80: How many bits does Extended ASCII use?

Extended ASCII uses 8 bits and includes a total of 256 characters, extra symbols, and diacritics.

Unicode, Files & Von Neumann Architecture

Question 81: What is the primary purpose of Unicode?

Unicode maps and represents all graphic characters of any writing system globally as a universal standard.

Question 82: What is UTF-8 and what are its features?

UTF-8 is a variable-length encoding using 1 to 4 bytes per character, maintaining full compatibility with ASCII.

Question 83: What is the binary representation of Urdu letter 'پ'?

The Urdu letter 'پ' has a specific Unicode, encoded according to Unicode standards.

Question 84: What is the difference between UTF-16 and UTF-32?

UTF-16 uses 2 or 4 bytes per character, whereas UTF-32 consistently uses 4 bytes per character.

Question 85: How are files stored in computers?

All files (images, audio, video, text) are stored on storage as binary data (sequences of 0s and 1s).

Question 86: Explain different storage devices (HDD, SSD, Cloud Storage).

HDD uses rotating disks to write data. SSD provides fast access via flash memory. Cloud Storage stores data on remote servers via internet.

Question 87: State basic data storage unit measurements.

1 Byte = 8 bits, 1 KB = 1024 Bytes, 1 MB = 1024 KB, 1 GB = 1024 MB, 1 TB = 1024 GB.

Unit 1 Exercise Short Questions

Question 1: Define a system and its basic components.

A system is a set of connected components achieving a specific purpose. Basic components include input, processing, output, and storage.

Question 2: Name the basic components of Von Neumann Architecture.

Von Neumann Architecture has four basic components: 1- Memory, 2- CPU, 3- Input devices, 4- Output devices.

Question 3: Name the four key stages of Von Neumann instruction cycle.

The four key stages are: 1- Fetching, 2- Decoding, 3- Execution, 4- Storage.

Question 4: What is a key advantage of Von Neumann Architecture?

A key advantage is the Stored Program Concept, where data and program instructions are stored in the same memory.

Question 5: What is the primary purpose of ASCII encoding?

ASCII provides a standard numeric code for English letters, numbers, and symbols to enable accurate data exchange among computers.

Question 6: Why was Unicode needed?

Since ASCII only supported English, Unicode was needed to support millions of characters across all world languages (Urdu, Arabic, Chinese, etc.).

Question 7: Explain the effects of changing bit count on integer value.

Increasing bit count expands the storage range of integer values. E.g., 8-bit max is 255, while 16-bit max is 65,535.

Question 8: Explain system software and provide two examples.

System software controls and manages basic hardware components. Examples: Operating system (Windows) and device drivers.

Question 9: Differentiate between system software and application software.

System software controls hardware and provides a platform, whereas application software is designed for specific user tasks.

Question 10: Solve the following binary addition: $1100_2 + 1011_2$

Answer:

$$\begin{array}{r} 1\ 1\ 0\ 0_2 \\ +\ 1\ 0\ 1\ 1_2 \\ \hline 1\ 0\ 1\ 1\ 1_2 \end{array}$$

Question 11: Solve the following binary subtraction: $1010_2 - 0011_2$

Answer:

$$\begin{array}{r} 1\ 0\ 1\ 0_2 \\ -\ 0\ 0\ 1\ 1_2 \\ \hline 0\ 1\ 1\ 1_2 \end{array}$$

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